Quiz Game Project Documentation

Overview

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This document provides a comprehensive explanation of the enhanced console-based Quiz Game implemented in Python. The game features multiple categories, difficulty levels, timer functionality, and an advanced scoring system.

Project Structure

The project consists of the following files:

1. main.py - Main game logic and entry point
2. questions.py - Question database organized by categories and difficulty levels
3. utils.py - Utility functions for colors, animations, formatting, and timer functionality
4. scores.txt - Stores player scores (automatically created)
5. README.md - User guide and project information

Enhanced Features

1. Difficulty Levels

The quiz game now includes three distinct difficulty levels:

- Easy: 10 points per question, 15 seconds per question
- Medium: 20 points per question, 12 seconds per question
- Hard: 30 points per question, 10 seconds per question

Additionally, there is an Adaptive Difficulty mode that automatically adjusts the question difficulty based on player performance:

- After 3 consecutive correct answers, the game presents harder questions
- After 2 consecutive incorrect answers, the game presents easier questions
- Otherwise, the game maintains medium difficulty

2. Timer Functionality

Each question has a time limit based on its difficulty level:

- Time limits provide a challenging yet fair gameplay experience
- Visual countdown timer shows remaining time for each question
- Players earn bonus points for answering quickly
- If time expires before answering, the question is marked incorrect

Bonus points for quick answers are calculated as:

- Base bonus varies by difficulty (2 for Easy, 5 for Medium, 10 for Hard)
- Actual bonus = Base bonus Ã— (Time remaining / Total time) Ã— 2

3. Enhanced Scoring System

The scoring system has been significantly improved with multiple ways to earn points:

a) Base Points:

- Easy questions: 10 points
- Medium questions: 20 points
- Hard questions: 30 points

b) Streak Bonuses:

- Players earn 2 bonus points for each consecutive correct answer
- Streak resets when a question is answered incorrectly

c) Time-Based Bonuses:

- Up to 10 bonus points for answering quickly
- More points for faster responses

d) Penalties:

- Half points deducted for incorrect answers (except on Easy difficulty)
- Encourages thoughtful responses while maintaining accessibility for beginners

e) Multipliers:

- Higher base points for harder questions
- Risk-reward balance encourages strategic play

Game Flow

1. Player enters their name
2. Selects a category (specific or mixed)
3. Chooses difficulty mode (Easy, Medium, Hard, or Adaptive)
4. Questions are presented with visual timer
5. Player answers before time expires
6. Immediate feedback with points calculation
7. Final score display with performance evaluation
8. Score saved to file with timestamp
9. Option to play again or exit

Technical Implementation

- Modular design with separate files for questions, utilities, and main logic
- Colorful console output using ANSI escape codes
- Emojis for enhanced visual appeal
- Typing animations for better user experience
- Threading for non-blocking input with timeout
- File I/O for persistent score storage
- Cross-platform compatibility (Windows/macOS/Linux)

User Experience Features

- Clean, intuitive interface with clear instructions
- Visual indicators for difficulty levels and points
- Performance feedback with emojis and messages
- High scores leaderboard
- Responsive design with smooth animations

- Comprehensive error handling

This enhanced quiz game provides a challenging and engaging experience that adapts to each player's skill level while maintaining an attractive visual presentation in the console environment.